

Arbitrage Pricing Theory (APT) Formula and How It's Used

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What Is the Arbitrage Pricing Theory (APT)?

Arbitrage pricing theory (APT) is a multi-factor asset pricing model. It's based on the idea that an asset's returns can be predicted using the linear relationship between the asset's expected return and a number of macroeconomic variables that capture systematic risk. It is a useful tool for analyzing portfolios from a value investing perspective, in order to identify securities that may be temporarily mispriced.

Key Takeaways

- Arbitrage pricing theory (APT) is a multi-factor asset pricing model based on the idea that an asset's returns can be predicted using the linear relationship between the asset's expected return and a number of macroeconomic variables that capture systematic risk.
- Unlike the CAPM, which assumes markets are perfectly efficient, APT assumes markets sometimes misprice securities, before the market eventually corrects and securities move back to fair value.
- Using APT, arbitrageurs hope to take advantage of any deviations from fair market value.

The Formula for the Arbitrage Pricing Theory Model Is

$$E(R)_i = E(R)_z + (E(I) - E(R)_z) \times \beta_i$$
where: $E(R)_i$ = Expected return on the asset
 R_z = Risk-free rate of return
 β_i = Sensitivity of the asset price to macroeconomic factor n
 E_i = Risk premium associated with factor i

The beta coefficients in the APT model are estimated by using linear regression. In general, historical securities returns are regressed on the factor to estimate its beta.

How the Arbitrage Pricing Theory Works

The arbitrage pricing theory was developed by the economist Stephen Ross in 1976, as an alternative to the capital asset pricing model (CAPM).¹ Unlike the CAPM, which assume markets are perfectly efficient, APT assumes markets sometimes misprice securities, before the market eventually corrects and securities move back to fair value. Using APT, arbitrageurs hope to take advantage of any deviations from fair market value.

However, this is not a risk-free operation in the classic sense of arbitrage, because investors are assuming that the model is correct and making directional trades—rather than locking in risk-free profits.

Mathematical Model for the APT

While APT is more flexible than the CAPM, it is more complex. The CAPM only takes into account one factor—market risk—while the APT formula has multiple factors. And it takes a considerable amount of research to determine how sensitive a security is to various macroeconomic risks.

The factors as well as how many of them are used are subjective choices, which means investors will have varying results depending on their choice. However, four or five factors will usually explain most of a security's return. (For more on the differences between the CAPM and APT, read more about how [CAPM and arbitrage pricing theory differ](#).)

APT factors are the systematic risk that cannot be reduced by the diversification of an investment portfolio. The macroeconomic factors that have proven most reliable as price predictors include unexpected changes in inflation, [gross national product](#) (GNP), corporate bond spreads and shifts in the yield curve. Other commonly used factors are [gross domestic product](#) (GDP), commodities prices, market indices, and exchange rates.

Example of How Arbitrage Pricing Theory Is Used

For example, the following four factors have been identified as explaining a stock's return and its sensitivity to each factor and the risk premium associated with each factor have been calculated:

- Gross domestic product (GDP) growth: $\beta = 0.6$, $RP = 4\%$
- Inflation rate: $\beta = 0.8$, $RP = 2\%$
- Gold prices: $\beta = -0.7$, $RP = 5\%$
- Standard and Poor's 500 index return: $\beta = 1.3$, $RP = 9\%$
- The risk-free rate is 3%

Using the APT formula, the expected return is calculated as:

$$\text{Expected return} = 3\% + (0.6 \times 4\%) + (0.8 \times 2\%) + (-0.7 \times 5\%) + (1.3 \times 9\%) = 15.2\%$$

What Is the Difference Between CAPM and Arbitrage Pricing Theory?

The main difference is that while CAPM is a single-factor model, the APT is a multi-factor model. In the CAPM, the only factor considered to explain the changes in the security prices and returns is the market risk. In the APT, on the other hand, the factors can be several.

What Are the Limitations of APT?

The main limitation of APT is that the theory does not suggest factors for a particular stock or asset. One stock could be more sensitive to one factor than another, and investors have to be able to perceive the risk sources and sensitivities.

What Is the Main Advantage of APT?

The main advantage of APT is that it allows investors to customize their research since it provides more data and it can suggest multiple sources of asset risks.

The Bottom Line

Arbitrage pricing theory (APT) is a multi-factor asset pricing model based on the idea that an asset's returns can be forecasted with the linear relationship between an asset's return and a number of macroeconomic factors that affect the asset's risk.